CMI: Cluster-Centric Missing Value Imputation with Feature Consistency

Megha Gupta
Department of Applied Data Science
San Jose State University
San Jose, USA
megha.gupta@sjsu.edu

Shripal Shah
Department of Applied Data Science
San Jose State University
San Jose, USA
shripal.shah@sjsu.edu

Mohammad Masum*
Department of Applied Data Science
San Jose State University
San Jose, USA
mohammad.masum@sjsu.edu

Sai Chandra Kosaraju
University of Nevada
Nevada, USA
sai.kosaraju@unlv.edu

2024 IEEE 14th Annual Computing and Communication Workshop and Conference (CCWC) | 979-8-3503-6013-4/24/$31.00 ©2024 IEEE | DOI: 10.1109/CCWC60891.2024.10427575

***Abstract*— In the realm of data analysis, addressing missing data poses a critical challenge with implications for both research and practical applications. The absence of data points in datasets can significantly undermine the reliability and performance of predictive models, potentially leading to erroneous conclusions. This paper introduces a novel approach, Cluster-Centric Missing Value Imputation (CMI), designed specifically for imputing missing values in numerical features using clustering techniques. CMI is augmented by Shapley Additive Explanations (SHAP) values to interpret feature significance post-imputation. The core principle of CMI lies in recognizing that data points within the same cluster often share similar key attributes, enhancing the transparency and understandability of the imputation process. Experimental evaluation on two medical datasets, the Indian Liver Patient Dataset (ILPD) and Chronic Kidney Disease Data (CKD), demonstrates the superior performance and interpretability of CMI compared to traditional imputation methods such as mean imputation, k-nearest neighbors (KNN) imputation, and Multiple Imputation by Chained Equations (MICE). The findings suggest that CMI represents a significant advancement in data analysis, providing an effective and interpretable solution for handling missing data in healthcare research.**

Keywords*— *Missing Value Imputation, Clustering, SHAP Values, Feature Important Analysis, Healthcare Data Analysis.

I. INTRODUCTION

In the domain of data analysis, the presence of missing values in datasets is a pervasive and challenging issue. A missing value occurs when there is no stored data value for a variable in an observation. These missing values can significantly compromise the performance and reliability of predictive models, leading to misleading conclusions. One way to get rid of these missing values is to use the imputation technique. Imputation is a technique for replacing missing data with substituted values. Traditional imputation methods, such as mean imputation, K-Nearest Neighbors (KNN), random sampling, and Multiple Imputation by Chained Equations (MICE), offer some solutions, but often at the cost of oversimplifying the complexity inherent in medical data [1].

Each imputation technique has its own limitations. Mean imputation, which involves replacing missing values with the average of observed values in the same variable, often leads to an overestimation of data variance and overlooks underlying data trends [1]. Random imputation, the process of substituting missing values with randomly selected observed values from the same variable, is easy to implement but can distort the original data distribution and neglect relationships between variables [22]. KNN, based on the proximity of data points and imputing missing values using the 'k' nearest observed values, struggles in high-dimensional spaces and is sensitive to the selection of number of nearest neighbors, and various distance metric used in distance calculation between two observations [1]. MICE, a more advanced strategy, imputes missing values through a series of regression models across multiple imputations. It assumes that data is missing at random and can be complex in model specification and interpretation due to its generation of multiple imputed datasets [8].

While these methods are useful in certain contexts, they highlight the trade-offs between simplicity, computational demands, and the preservation of original data characteristics in missing data imputation. In addition, existing imputation techniques lack consistency in providing feature importance. Therefore, in this paper, we introduce a novel framework, Cluster-Centric Missing Value Imputation (CMI), based on clustering approach to impute missing values that can offer comparable performance in imputing missing values while ensuring consistent feature importance ranking. Clustering enables the identification of inherent patterns and relationships within the data, providing a robust framework for predicting missing values [2]. We apply SHAP (SHapley Additive explanations) values, a method for explaining the contribution of each feature in prediction models, to assess feature importance [4, 5]. To demonstrate the effectiveness, we apply our proposed framework on two datasets from the medical domain. This choice is motivated by the critical importance of accurate data analysis in healthcare research, where missing values present unique challenges and necessitate advanced imputation techniques with proper feature importance ranking tailored for informed decision-making and improved patient care. Experimental results show that CMI not only provides approximately equivalent performance for missing value imputation but also

* Mohammad Masum is the corresponding author

979-8-3503-6013-4/24/$31.00 ©2024 IEEE

thorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on July 24,2026 at 04:26:21 UTC from IEEE Xplore. Restrictions appl

enhances feature importance ranking, reinforcing its suitability for robust healthcare data analysis.

II. Related Work

A comparative analysis of five missing data imputation techniques (Mean/Mode, K-Nearest Neighbor, Hot-Deck, Expectation Maximization, and C5.0) is presented, emphasizing critical factors influencing method selection [1]. The research introduces an innovative framework deviating from standard imputation approaches, employing random filling and a Binary Cluster-Association matrix (BA) for clustering. This approach demonstrates superior cluster accuracy with potential applications in cybersecurity, showcasing advancements in handling missing values [2]. The exploration extends to clustering algorithms' adaptability to datasets with missing values, underscoring the necessity for evolving techniques, particularly in bioinformatics [3]. Furthermore, a hybrid imputation approach integrating single and multiple techniques exhibits notable enhancements in imputation accuracy [6]. In the healthcare domain, the study addresses missing data challenges, proposing three novel imputation techniques leveraging unique correlation approaches, surpassing existing methods in precision and classification performance [7]. Additionally, a novel framework is presented, enhancing imputation accuracy without escalating computational complexity, marking significant progress in managing missing values in large-scale databases [8]. Contrary to earlier beliefs, the research highlights attainable unbiased results in data analysis with up to 90% missing data, emphasizing the significance of correctly specified imputation models and random missing data [9]. This study challenges conventional wisdom, directing attention from the proportion to the fraction of missing information for methodological decisions in data analysis.

III. Methodology

Figure 1 illustrates the proposed framework's flowchart. In the first step, datasets are selected. Subsequently, we add varying levels of missing data, specifically at 20%, 40%, 60%, and 80%, to simulate different scenarios of data deficiency. This involves the synthetic introduction of missing values in a random manner across the dataset. To achieve this, we randomly select a proportionate number of data points at each incremental level and replace them with NaN (Not a Number) values. As the CMI technique is primarily designed for imputing numerical missing values, we address categorical features at the outset of the process. Thereafter, we convert categorical attributes into numerical representations. This comprehensive approach is consistently applied across all five imputation techniques, including CMI.

The CMI's block diagram is shown in Figure 2. The data is sourced from the UCI Machine Learning Repository. We address categorical values by employing mode imputation, replacing them with the most frequently occurring category to ensure data integrity and consistency [1]. In the later step, we standardize the dataset excluding the target variable. For numerical data, we initially replace missing values with a randomly selected constant number within the z-score range of (-3, 3), corresponding to three standard deviations from the mean in a standardized dataset.

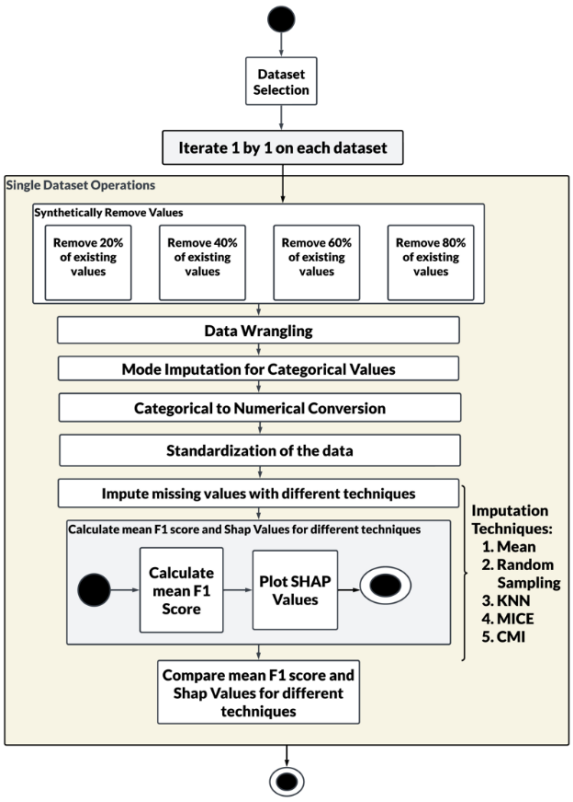


Figure 1: Flowchart of the Experiment for Imputation of Missing Values Using Various Imputation Techniques

This preliminary step is necessary because K-means clustering, which forms the basis of our methodology, does not perform optimally with missing values. Filling these gaps with a constant ensures the algorithm can effectively analyze the entire dataset. We then apply the elbow method to determine the optimal number of clusters and utilize k-means clustering to segment the dataset into distinct clusters [23].

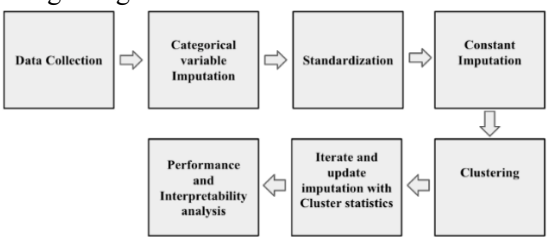


Figure 2: Block Diagram Depicting the CMI Framework to Impute Missing Values via Clustering

By considering the inherent structure and patterns within the data, CMI aims to provide more contextually accurate imputations compared to traditional methods. By calculating the means of each cluster, we iteratively impute missing values with these means. In the next step, we apply a Machine Learning classifier (Random Forest) and perform a comparative analysis to assess the effectiveness of our proposed framework. Alongside this, SHAP values are analyzed for interpretability and to define the contribution of individual features to the model's predictions. Experiment and Results

thorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on July 24,2026 at 04:26:21 UTC from IEEE Xplore. Restrictions appl

A. Dataset Specification

We utilized two datasets from the UCI Machine Learning Repository: Indian Liver Patient Dataset (ILPD) and the chronic kidney disease (CKD) dataset. Both datasets are multivariate and are employed for binary classification tasks. The ILPD consists of 583 instances with 10 variables such as Age, Gender, Total Bilirubin (TB), Direct Bilirubin (DB), Alkphos Alkaline Phosphatase (Phosphotase), Total Protein (TP), Albumin (ALB), A/G Ratio, Alamine Aminotransferase (SGPT), Aspartate Aminotransferase (SGOT) [20]. This dataset includes records from 416 patients diagnosed with liver disease and 167 without having liver disease. In contrast, the CKD dataset contains 400 instances with 25 features, offering a comprehensive view of various medical parameters relevant to kidney disease. It comprises 400 instances and 25 features, which include a mix of 11 numeric and 14 nominal attributes such as Specific Gravity (SG), Albumin (AL), Blood Urea (BU), Serum Creatinine (SC), Hemoglobin (HB), Packed Cell Volume (PCV), Hypertension (HT) [21]. These attributes encompass various medical parameters like age, blood pressure, specific gravity, albumin, sugar levels, and more, along with a binary classification of chronic kidney disease (CKD/not CKD).

B. Data Preprocessing

In the Indian Liver Patient Dataset (ILPD), we first analyze the class distribution to assess the balance of the dataset. This analysis is followed by a bar chart, as illustrated in Figure 3, which quantifies the missing data across different features, highlighting the proportion of missing data in each column. The data cleaning process addresses the categorical features, such as 'Gender' in this dataset, through mode imputation. During the data transformation phase, the categorical variable 'Gender' is subjected to Label Encoding, converting it into a numeric format. Additionally, dataset is standardized using the StandardScaler, ensuring a consistent scale across the dataset.

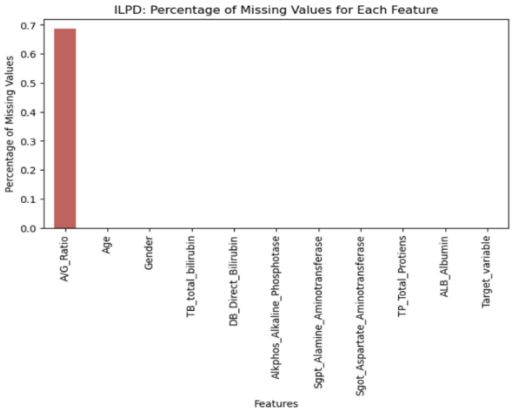


Figure 3: Graphical Representation of Missing Values in Each Feature of the Indian Liver Patient Dataset

Simultaneously, the CKD dataset undergoes a similar yet distinct preprocessing approach. A graphical representation of missing values was created and displayed in Figure 4 for each feature, using a bar chart.

The data exploration step involves removing the non-essential 'id' column as this column's values are arbitrary and do not contribute to our analysis or predictive modelling. During the data cleaning process, we focus on converting columns that were originally in text (object) format into numeric representations. This stage is critical for resolving data type discrepancies, which is required for future analysis and modelling. Ambiguities in categorical data, particularly in columns like diabetes_mellitus, coronary_artery_disease, and class, are resolved through manual corrections. The data transformation for the CKD dataset includes Label Encoding for all categorical columns. Numeric features are standardized to normalize the data, ensuring uniformity in subsequent analyses.

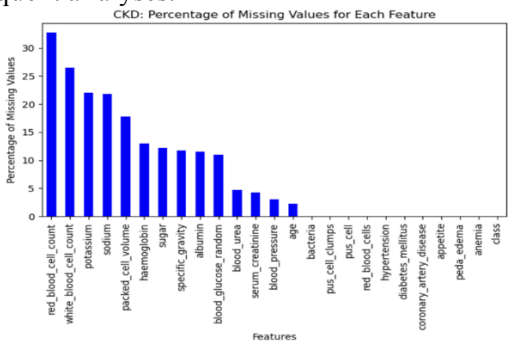


Figure 4: Graphical Representation of Missing Values in Each Feature of the Chronic Kidney Disease Dataset

C. Model Evaluation Metrics

In assessing our model's performance on imbalanced datasets, we prioritize the F1 score (eq. 1) as a comprehensive metric, capturing both precision and recall [19]. We conducted 30 repetitions for each experiment, reporting the mean F1 score with the Standard Deviation (SD). The mean F1 score offers an average performance measure, and the SD provides insights into prediction consistency across datasets.

$$F1\ Score = \frac{2 * Precision * recall}{Precision + recall} \qquad \text{eq. 1}$$

D. Experimental Design

We conducted four experiments introducing varying levels of missing values (20%, 40%, 60%, and 80%) across datasets, evaluating the proposed CMI technique against Mean, KNN, Random, and MICE imputation methods. To establish baseline features for liver and kidney disease patients, we referenced medical journals, identifying the top four features via SHAP values. Hyperparameters for the Random Forest classifier was set with 10 estimators, employing an 80-20 stratified train-test split to maintain class ratios. Data standardization, preventing data leakage, utilized training data statistics. In CMI, we performed an initial random imputation within a range of -3 to 3, employed KMeans clustering with four clusters using the Elbow method, and ensured consistency and reproducibility in cluster examination for the ILPD and CKD datasets. Fig. 5 and 6 illustrate the cluster distribution using PCA for ILPD, and CKD datasets, respectively. Cluster centroids, marked as black 'X', denoted mean positions of data points, guiding iterative imputation based on the cluster mean for handling missing values.

thorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on July 24,2026 at 04:26:21 UTC from IEEE Xplore. Restrictions appl

E. Experimental Results

For a comprehensive assessment of performance across different scenarios, the mean F1 score was calculated 30 times for each level of missing data (20%, 40%, 60%, and 80%), and SHAP values were plotted for each imputation method. The ranking of features was indicated by colors while comparing SHAP values among different imputation techniques, as shown in Tables 2 to 10 (except Table 6): red for the most important feature, blue for the second, green for the third, purple for the fourth, and grey for all other features of lesser importance identified by the models.

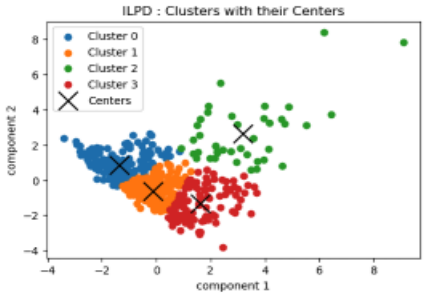


Figure 5: Cluster Distribution of ILPD Dataset with Centroid Markings Using K-Means Clustering

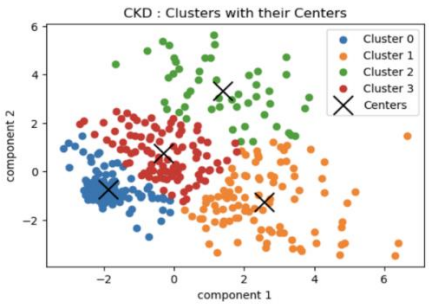


Figure 6: Cluster Distribution of CKD Dataset with Centroid Markings Using K-Means Clustering

Table 1: F1 Scores for different rates of Missing Values in the ILPD Dataset using different mean imputation techniques.

Missing Values Percentages	F1 Score (STD) for each Imputation Technique				
	Mean Imputation	Random Sampling	KNN Imputation	MICE	CMI
20%	0.783 (0.03)	0.776 (0.03)	0.796 (0.02)	0.781 (0.03)	0.793 (0.02)
40%	0.784 (0.03)	0.784 (0.03)	0.788 (0.03)	0.779 (0.03)	0.785 (0.02)
60%	0.775 (0.03)	0.794 (0.03)	0.782 (0.03)	0.786 (0.03)	0.793 (0.02)
80%	0.779 (0.02)	0.777 (0.03)	0.776 (0.02)	0.771 (0.03)	0.789 (0.02)

In evaluating liver disease markers, Total Bilirubin (TB), Direct Bilirubin (DB), Alkphos Alkaline Phosphatase (Phosphatase), and Sgpt Alamine Aminotransferase (SGPT) are identified as crucial indicators, with elevated TB and DB levels signifying liver dysfunction and cirrhosis [10] [11]. This study compares these features across five imputation techniques, utilizing SHAP values for interpretability and performance efficacy assessment [12, 13]. Table 1 illustrates F1 scores for these techniques at various missing value percentages, revealing CMI's robustness, outperforming Mean Imputation and MICE across scenarios [11].

Table 2, Table 3, Table 4, and Table 5 present CMI's performance at 20%, 40%, 60%, and 80% missing data, respectively. CMI consistently identifies the top four features—TB, DB, Phosphatase, and SGPT—matching medical journal findings and surpassing other methods in feature recognition and imputation accuracy [10, 11, 12, 13]. Even with 80% missing data, CMI maintains precise feature identification and ranking, highlighting its superior performance in handling substantial missing information.

Table 2: Comparison of Important Features Using SHAP Values in the ILPD Dataset at 20% Missing Values

Imp_Feature Ranks	Mean	Random	KNN	MICE	CMI
TB	SGOT	PH	SGOT	PH	PH
DB	DB	DB	DB	SGOT	TB
PH	PH	SGOT	PH	DB	SGPT
SGPT	TB	SGPT	AGE	AGE	DB

Table 3: Comparison of Important Features Using SHAP Values in the ILPD Dataset at 40% Missing Values

Imp_Feature Ranks	Mean	Random	KNN	MICE	CMI
TB	PH	TB	SGOT	SGOT	TB
DB	SGOT	PH	TB	DB	SGOT
PH	TB	SGOT	PH	SGPT	PH
SGPT	SGPT	AGE	AGE	AGE	SGPT

Table 4: Comparison of Important Features Using SHAP Values in the ILPD Dataset at 60% Missing Values

Imp_Feature Ranks	Mean	Random	KNN	MICE	CMI
TB	SGOT	PH	PH	PH	TB
DB	DB	DB	SGPT	AGE	PH
PH	TB	SGOT	TB	DB	SGPT
SGPT	PH	AGE	SGOT	SGOT	SGOT

In CKD, four key health indicators are crucial identified by medical journals: Specific Gravity (SG), Albumin (AL), Blood Urea (BU), and Serum Creatinine (SC) [14, 15, 16, 17]. Our study examined and compared these CKD indicators with features identified by five different imputation methods, interpreted through SHAP values.

thorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on July 24,2026 at 04:26:21 UTC from IEEE Xplore. Restrictions appl

Table 5: Comparison of Important Features Using SHAP Values in the ILPD Dataset at 80% Missing Values

Imp_Feature Ranks	Mean	Random	KNN	MICE	CMI
TB	DB	SGPT	SGOT	SGPT	TB
DB	SGPT	PH	PH	TB	SGPT
PH	SGOT	SGOT	SGPT	ALB	PH
SGPT	PH	AGE	TB	PH	SGOT

Table 6: F1 Scores for different Missing Values in the CKD Dataset using different Imputation Techniques.

Missing Values Percentages	F1 Score (STD) for each Imputation Technique				
	Mean Imputation	Random Sampling	KNN Imputation	MICE	CMI
20%	0.973 (0.020)	0.972 (0.021)	0.969 (0.018)	0.993 (0.011)	0.976 (0.019)
40%	0.978 (0.016)	0.962 (0.023)	0.974 (0.020)	0.992 (0.011)	0.977 (0.025)
60%	0.971 (0.022)	0.971 (0.017)	0.974 (0.01)	0.992 (0.010)	0.976 (0.02)
80%	0.970 (0.023)	0.970 (0.02)	0.969 (0.021)	0.990 (0.014)	0.974 (0.023)

Table 6 presents the F1 scores for all five imputation techniques, accounting for varying percentages of missing values in chronic kidney disease (CKD) data. CMI demonstrates consistently strong performance across all levels of missing data, outshining other imputation methods. Specifically, CMI outperforms Mean Imputation, Random Sampling, KNN Imputation, and MICE in scenarios with 20% and 40% missing values. At higher levels of missing data, specifically 60% and 80%, CMI maintains performance comparable to the MICE method. This indicates its effectiveness and reliability in handling various contexts of missing data.

Table 7: Comparison of Important Features Using SHAP Values in the CKD Dataset at 20% Missing Values

Imp_Feature Ranks	Mean	Random	KNN	MICE	CMI
SG	PCV	SG	SG	HB	SG
AL	HB	HB	HB	SG	AL
BU	SG	SC	AL	SC	SC
SC	SC	AL	PCV	AL	HT

In Table 7, with 20% missing values, CMI skillfully identifies three key features: SG, AL, and SC, in alignment with the findings [14,15]. In contrast, other techniques such as Mean, Random, and KNN only manage to identify two of these features. Although MICE detect all three, it fails to maintain the order of importance.

Table 8: Comparison of Important Features Using SHAP Values in the CKD Dataset at 40% Missing Values

Imp_Feature Ranks	Mean	Random	KNN	MICE	CMI
SG	PCV	SG	SG	HB	SG
AL	HB	HB	HB	SG	AL
BU	SG	SC	AL	SC	SC
SC	SC	AL	PCV	AL	HT

In the analysis presented in Table 8, which focuses on a dataset with 40% missing data, CMI accurately identifies SG, AL, and SC as significant features. In comparison, other methods such as Mean, Random, and KNN succeed in recognizing only two of these features. While MICE identify all three, it does not accurately reflect the established order of importance.

Table 9: Comparison of Important Features Using SHAP Values in the CKD Dataset at 60% Missing Values

Imp_Feature Ranks	Mean	Random	KNN	MICE	CMI
SG	SC	HB	HB	HB	PCV
AL	PCV	SG	SC	SG	SG
BU	SG	RBC	SG	AL	SC
SC	HB	PCV	PCV	PCV	AL

In Table 9, when analyzing a dataset with 60% missing values, CMI effectively pinpoints SG (Specific Gravity), SC (Serum Creatinine), and AL (Albumin) as key features. In comparison, Mean, KNN, and MICE detect only two significant features, while Random identifies just one.

Table 10: Comparison of Important Features Using SHAP Values in the CKD Dataset at 80% Missing Values

Imp_Feature Ranks	Mean	Random	KNN	MICE	CMI
SG	HB	HB	HB	SC	SG
AL	PCV	SG	AL	PCV	SC
BU	SC	AL	SC	SG	PCV
SC	SG	PCV	RBC	HB	AL

thorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on July 24,2026 at 04:26:21 UTC from IEEE Xplore. Restrictions appl

In Table 10, which deals with a high rate of 80% missing data, the CMI method stands out by accurately identifying SG (Specific Gravity), AL (Albumin), and SC (Serum Creatinine) as critical features [16]. Other techniques, such as Mean, Random, KNN, and MICE, fall short in this regard, correctly identifying only two of these important features. Overall, from table 7-10, CMI stands out by not only recognizing the key features but also maintaining their priority as documented in medical research, further demonstrating its superior performance in imputation.

IV. Conclusion

In conclusion, our novel clustering-based imputation technique, CMI, demonstrates significant efficacy in handling missing values in datasets, particularly in the medical domain. Thoroughly tested on the ILPD and CKD datasets across various missing value percentages, CMI consistently outperforms traditional imputation methods, such as mean, KNN, random, and MICE, by effectively identifying critical disease-contributing features. Notably, CMI exhibits comparable results in most scenarios and surpasses existing methods in certain cases. Its strength lies in interpretability, as evidenced by SHAP values, providing accurate and clinically relevant data imputation. This underscores the potential of CMI as a robust tool for improving model performance, feature importance analysis, and decision-making in medical datasets with missing values.

Acknowledgment

We would like to express our sincere thanks to Gaurav Gupta for his help in reviewing the paper.

References

[1] T. Aljuaid and S. Sasi, "Proper imputation techniques for missing values in data sets," 2016 International Conference on Data Science and Engineering (ICDSE), Cochin, India, 2016, pp. 1-5, doi: 10.1109/ICDSE.2016.7823957.

[2] M. Pattanodom, N. Iam-On and T. Boongoen, "Clustering data with the presence of missing values by ensemble approach," 2016 Second Asian Conference on Defence Technology (ACDT), Chiang Mai, Thailand, 2016, pp. 151-156, doi: 10.1109/ACDT.2016.7437660.

[3] L. Himmelspach and S. Conrad, "Clustering approaches for data with missing values: Comparison and evaluation," 2010 Fifth International Conference on Digital Information Management (ICDIM), Thunder Bay, ON, Canada, 2010, pp. 19-28, doi: 10.1109/ICDIM.2010.5664691.

[4] Y. Shirota, M. Fujimaki, E. Tsujiura, M. Morita and J. A. D. Machuca, "A SHAP Value-Based Approach to Stock Price Evaluation of Manufacturing Companies," 2021 4th International Conference on Artificial Intelligence for Industries (AI4I), Laguna Hills, CA, USA, 2021, pp. 75-78, doi: 10.1109/AI4I51902.2021.00026.

[5] Yong-Geon Lee, Jae-Young Oh, Dongsung Kim, and Gibak Kim, "SHAP Value-Based Feature Importance Analysis for Short-Term Load Forecasting," Journal of Electrical Engineering & Technology, vol. 18, no. 2, pp. 579-588, 2023

[6] S. I. Khan and A. S. M. L. Hoque, "SICE: an improved missing data imputation technique," Journal of Big Data, vol. 7, no. 1, p. 37, 2020.

[7] I. Curioso et al., "Addressing the curse of missing data in clinical contexts: A novel approach to correlation-based imputation," Journal of King Saud University - Computer and Information Sciences, vol. 35, no. 6, p. 101562, 2023. doi:10.1016/j.jksuci.2023.101562

[8] A. Farhangfar, L. A. Kurgan and W. Pedrycz, "A Novel Framework for Imputation of Missing Values in Databases," in IEEE Transactions on Systems, Man, and Cybernetics - Part A: Systems and Humans, vol. 37, no. 5, pp. 692-709, Sept. 2007, doi: 10.1109/TSMCA.2007.902631.

[9] P. Madley-Dowd, R. Hughes, K. Tilling, and J. Heron, "The proportion of missing data should not be used to guide decisions on multiple imputation," Journal of Clinical Epidemiology, vol. 110, pp. 63–73, 2019. doi:10.1016/j.jclinepi.2019.02.016

[10] R. N. Das, "Determinants of total bilirubin for liver patients," J Liver Res Disord Ther, vol. 4, no. 3, pp. 130-134, Jun. 2018. doi: 10.15406/jlrdt.2018.04.00115.

[11] A. R. Guerra Ruiz, J. Crespo, R. M. López Martínez, P. Iruzubieta, G. Casals Mercadal, M. Lalana Garcés, B. Lavin, and M. Morales Ruiz, "Measurement and clinical usefulness of bilirubin in liver disease," Advances in Laboratory Medicine / Avances en Medicina de Laboratorio, vol. 2, no. 1, pp. 1–10, Jul. 2021, doi: 10.1515/almed-2021-0047

[12] B. H. Al Telaq and N. Hewahi, "Prediction of Liver Disease using Machine Learning Models with PCA," 2021 International Conference on Data Analytics for Business and Industry (ICDABI), Sakheer, Bahrain, 2021, pp. 250-254, doi: 10.1109/ICDABI53623.2021.9655897.

[13] T. M. Ghazal, A. U. Rehman, M. Saleem, M. Ahmad, S. Ahmad and F. Mehmood, "Intelligent Model to Predict Early Liver Disease using Machine Learning Technique," 2022 International Conference on Business Analytics for Technology and Security (ICBATS), Dubai, United Arab Emirates, 2022, pp. 1-5, doi: 10.1109/ICBATS54253.2022.9758929.

[14] N. R. Silva et al., "Association of uric acid and uric acid to creatinine ratio with chronic kidney disease in hypertensive patients," BMC Nephrology, vol. 22, no. 1, p. 311, Sep. 2021, doi: 10.1186/s12882-021-02521-9

[15] Chen TK, Knicely DH, Grams ME. Chronic Kidney Disease Diagnosis and Management. JAMA. 2019;322(13):1294-1304. doi:10.1001/jama.2019.14745

[16] E. Dritsas and M. Trigka, "Machine Learning Techniques for Chronic Kidney Disease Risk Prediction," Big Data Cogn. Comput., vol. 6, no. 3, pp. 98, Oct. 2022, doi: 10.3390/bdcc6030098

[17] A. Farjana et al., "Predicting Chronic Kidney Disease Using Machine Learning Algorithms," 2023 IEEE 13th Annual Computing and Communication Workshop and Conference (CCWC), Las Vegas, NV, USA, 2023, pp. 1267-1271, doi: 10.1109/CCWC57344.2023.10099221.

[18] N. Karmitsa, S. Taheri, A. Bagirov and P. Mäkinen, "Missing Value Imputation via Clusterwise Linear Regression," in IEEE Transactions on Knowledge and Data Engineering, vol. 34, no. 4, pp. 1889-1901, 1 April 2022, doi: 10.1109/TKDE.2020.3001694.

[19] M. K. Warmuth, S. Hanneke, and J. Duchi, "F*: an interpretable transformation of the F-measure," Machine Learning, vol. 111, no. 2-3, pp. 357-373, Feb. 2022, doi: 10.1007/s10994-021-05964-1.

[20] B. Ramana and N. Venkateswarlu, "ILPD (Indian Liver Patient Dataset)," UCI Machine Learning Repository, 2012, https://doi.org/10.24432/C5D02C.

[21] L. Rubini, P. Soundarapandian, and P. Eswaran, "Chronic_Kidney_Disease," UCI Machine Learning Repository, 2015,https://doi.org/10.24432/C5G020.

[22] P. C. Chiu, A. Selamat, O. Krejcar, K. K. Kuok, S. D. A. Bujang and H. Fujita, "Missing Value Imputation Designs and Methods of Nature-Inspired Metaheuristic Techniques: A Systematic Review," in IEEE Access, vol. 10, pp. 61544-61566, 2022, doi: 10.1109/ACCESS.2022.3172319

[23] C.-H. Cheng and S.-F. Huang, A novel clustering-based purity and distance imputation for handling medical data with missing values, 2021. doi:10.21203/rs.3.rs-520996/v1

thorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on July 24,2026 at 04:26:21 UTC from IEEE Xplore. Restrictions appl